

Digital Filter Design for Narrowband Interference Rejection

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1 Problem Statement

In receiver front-ends, recovering a weak signal of interest (SOI) often requires filtering out both wide-band noise and strong narrowband interferers (such as adjacent-channel signals or jamming tones). This project models this receiver problem and designs two digital filters to isolate the SOI.

The test signal consists of:

- **Signal of interest (SOI):** 1000 Hz sine wave, unit amplitude
- **Narrowband interferer:** 3200 Hz sine wave at $3\times$ the SOI amplitude
- **Additive White Gaussian Noise (AWGN):** Set for an input SNR of -10.9 dB

Figure 1 shows the spectrum of the corrupted signal. The SOI sits near the noise floor, while the interferer is a strong peak at 3200 Hz.

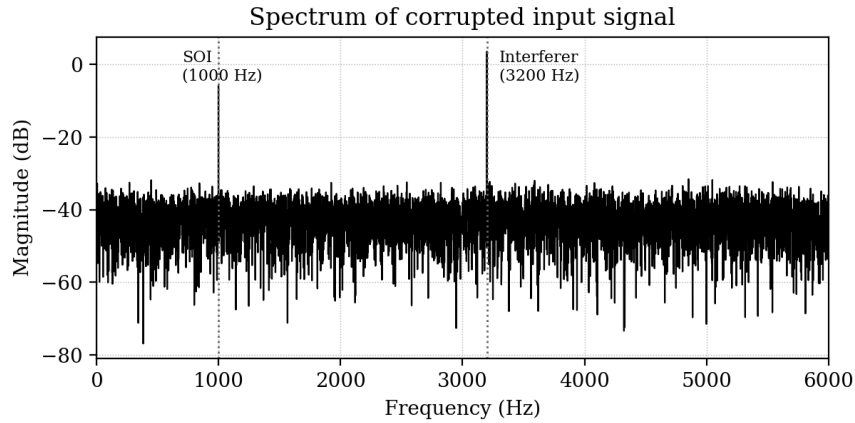


Figure 1: Spectrum of the corrupted input signal showing the weak 1000 Hz SOI and strong 3200 Hz interferer.

2 Filter Design & Specifications

Both FIR and IIR filters were designed to meet the same target filter specification:

Table 1: Digital Filter Design Specifications

Parameter	Specification Value
Passband Edge	1400 Hz
Stopband Edge	2600 Hz
Passband Ripple	1 dB
Stopband Attenuation	40 dB
Sampling Frequency (f_s)	20,000 Hz

2.1 FIR Design (Parks-McClellan / Equiripple)

The FIR filter was designed using the Parks-McClellan (Remez exchange) algorithm to yield an equiripple passband and stopband. The required filter order was estimated using Bellanger’s formula, resulting in a 23-tap Type-I linear-phase FIR filter.

2.2 IIR Design (Butterworth)

The IIR filter was designed using a Butterworth prototype via `scipy.signal.buttord`, resulting in an 8th-order filter (4 second-order sections in SOS form for numerical stability).

Figure 2 compares the magnitude responses of both designs. Both filters hit the 40 dB stopband attenuation spec, but the IIR filter does so with roughly $2.9\times$ fewer coefficients.

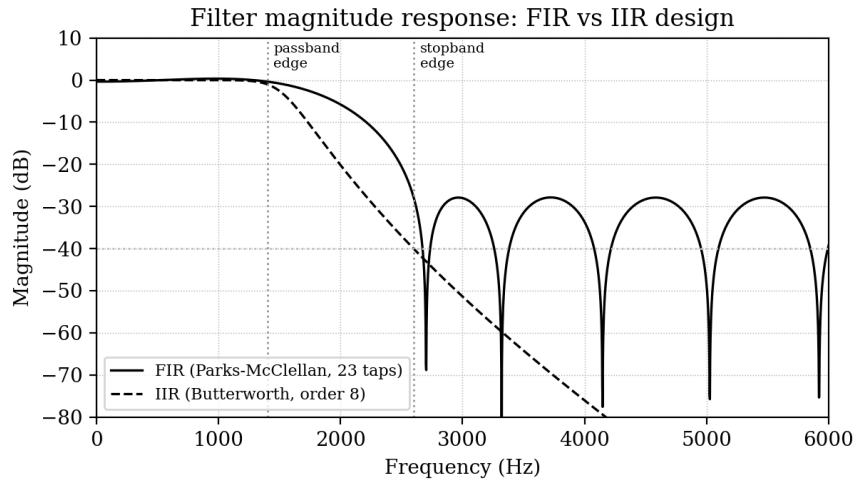


Figure 2: Magnitude response comparison between the 23-tap FIR filter and 8th-order Butterworth IIR filter.

3 Results & Trade-off Analysis

3.1 Group Delay

The FIR filter provides linear phase, yielding a constant group delay of 0.55 ms across the passband. The IIR filter has non-linear phase, with group delay varying up to 0.97 ms near the band edge (Figure 3). Constant group delay preserves signal waveform shape, while non-constant delay causes dispersion.

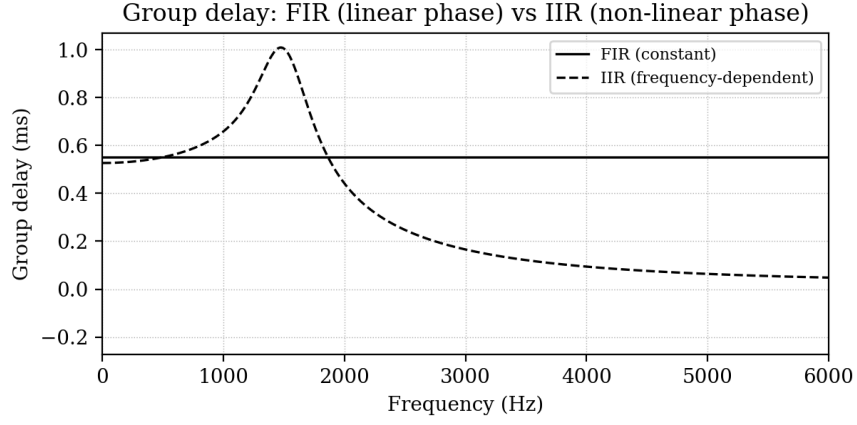


Figure 3: Group delay comparison showing linear phase (constant delay) for FIR vs. non-linear phase for IIR.

3.2 Interference Rejection and SNR

Table 2 summarizes the performance of both filters after aligning the constant FIR delay.

Table 2: Measured Filter Performance

Signal State	Filter Complexity	Output SNR	SNR Improvement
Unfiltered Input	–	–10.90 dB	–
FIR (Parks-McClellan)	23 taps	+2.23 dB	+13.14 dB
IIR (Butterworth)	8th order	–6.42 dB	+4.49 dB

Both filters suppress the 3200 Hz interferer to the noise floor (Figure 4). The FIR filter achieves a higher SNR improvement because its linear phase preserves the SOI waveform exact alignment, whereas the IIR filter’s phase distortion introduces slight shape error relative to the clean reference tone.

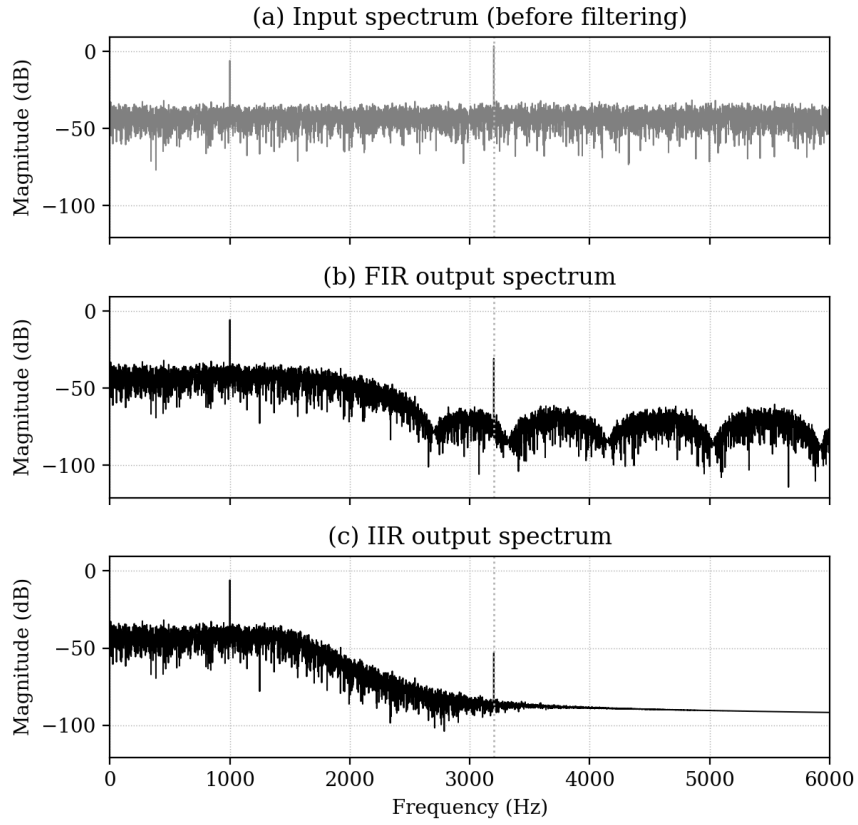


Figure 4: Spectrum before and after filtering, showing complete removal of the 3200 Hz interferer tone.

4 Summary

Both the 23-tap FIR filter and 8th-order IIR filter successfully reject the strong narrowband interferer. The IIR filter achieves the spec with $3\times$ fewer coefficients, making it computationally efficient for real-time systems. However, the FIR filter provides linear phase, preserving signal shape without phase distortion.